PSRP Installation Guide

This guide covers deploying a PSRP (PRE Security SOAR Remote Presence) VM into a customer network.

Prerequisites

- A hypervisor capable of running QCOW2 images (QEMU/KVM, Proxmox, libvirt, or similar)
- The PSRP QCOW2 image for your target architecture (`psrp-amd64.qcow2` or `psrp-arm64.qcow2`)
- Network connectivity from the VM to the internet (outbound HTTPS/SSH)
- The DNS address of your PRE Security server (e.g. `server.presecurity.ai`)

The instructions that follow show how the ARM64 image can be run in an environement using QEMU. Any hypervisor can be used although the instructions for instantiation are outside the scope of this document.

After the VM is running, see the instructions below for connecting to the SSH and/or the browser interface for configuration steps. These will be the same regardless of which hypervisor framework is used.

Obtaining the image

These instructions are obtained by the command

```
curl https://sw-download.presecurity.ai:/psrp/INSTALL.pdf -o INSTALL.pdf
```

The following curl command can be used to obtain the ARM image

```
curl https://sw-download.presecurity.ai:/psrp/psrp-arm64.qcow2 -o psrp-arm64.qcow2
```

For the AMD image use

```
curl https://sw-download.presecurity.ai:/psrp/psrp-amd64.qcow2 -o psrp-amd64.qcow2
```

Step 1: Import the VM Image

Import the QCOW2 image into your hypervisor. The VM requires:

Resource	Minimum	Recommended
RAM	1 GB	2 GB
CPU	1 vCPU	2 vCPUs
Disk	Included in image	—
Network	1 NIC, bridged or NAT with outbound access	—

QEMU/KVM Example

```
qemu-system-x86_64 \
    -machine q35 \
    -cpu host -accel kvm \
    -m 2048 -smp 2 \
```

PSRP Configuration

Figure 1: PSRP Configuration

```
    -drive file=psrp-amd64.qcow2,format=qcow2,if=virtio \
    -netdev user,id=net0,hostfwd=tcp::2222-:22,hostfwd=tcp::8080-:80 \
    -device virtio-net-pci,netdev=net0 \
    -display none -daemonize
```

This forwards SSH to host port 2222 and the configuration web UI to host port 8080. Adjust port forwarding or use bridged networking as appropriate for your environment.

ARM64

For ARM64 hosts, use `qemu-system-aarch64` with EFI firmware:

```
qemu-system-aarch64 \
    -machine virt \
    -cpu host -accel kvm \
    -m 2048 -smp 2 \
    -drive if=pflash,format=raw,unit=0,file=/usr/share/AAVMF/AAVMF_CODE.fd,readonly=on \
    -drive if=pflash,format=raw,unit=1,file=efivars.fd \
    -drive file=psrp-arm64.qcow2,format=qcow2,if=virtio \
    -netdev user,id=net0,hostfwd=tcp::2222-:22,hostfwd=tcp::8080-:80 \
    -device virtio-net-pci,netdev=net0 \
    -display none -daemonize
```

This command will start the VM.

Checking Status and Controlling the VM

If you have the source repository, use the included helper scripts:

```
./scripts/run_vm.sh                    # Start VM (snapshot mode, image not modified)
./scripts/run_vm.sh --persist          # Start VM (changes persist to disk)
./scripts/stop_vm.sh                   # Stop the VM
./scripts/status.sh                    # Check VM status
```

Step 2: Boot and Access the Configuration Page

Once the VM boots (typically 30-60 seconds), open a web browser and navigate to the VM's HTTP port:

```
http://<vm-host>:8080/
```

If using bridged networking, use the VM's IP address directly on port 80.

You will see the PSRP Configuration page:

Step 3: Configure the PSRP

Enter the following details:

- **PRE Server DNS Address** – The DNS address of the PRE Security server that this PSRP will connect to (e.g. `server.presecurity.ai`).
- **System Name** – A descriptive name to identify this PSRP instance on the PRE Server (e.g. `client-office-01`).

Click **Configure** to submit. The PSRP will:

1. Save the configuration locally
2. Begin establishing an outbound connection to the PRE Server
3. Redirect you to a status page showing the connection progress

Step 4: Verify Registration

After configuration, the status page will show the current connection state:

- **CONNECTING** – The PSRP is attempting to reach the PRE Server. This is normal for the first few minutes while the secure tunnel is established and the server approves the client.
- **PROVISIONED** – The PSRP has successfully connected and is operational. The MCP server is running and the PRE Server can issue security operations to this device.

On the PRE Server side, the PSRP should appear on the **Manage Remote Clients** page once the connection is established and approved.

Step 5: Network Considerations

The PSRP establishes an **outbound** SSH tunnel to the PRE Server. No inbound ports need to be opened on the customer firewall. Ensure the following outbound access is available:

Destination	Port	Protocol	Purpose
PRE Server DNS	443	HTTPS	Registration and API
PRE Server DNS	SSH tunnel port	TCP	Persistent tunnel for MCP operations

Troubleshooting

VM does not boot

- Verify the QCOW2 image is not corrupted (check file size against the release checksum)
- ARM64 images require EFI firmware – ensure pflash drives are configured
- Check that hardware acceleration is available (`kvm` on Linux, `hvf` on macOS)

Configuration page not loading

- Wait at least 60 seconds after boot for all services to start
- Verify port forwarding is correct (guest port 80 must be reachable)

- SSH into the VM and check: `systemctl status psrp-config-server`

PSRP stuck in CONNECTING state

- Verify the PRE Server DNS address is correct and reachable from the VM
- Check outbound network access: `ssh -p 2222 psrp@<host>` then `curl -v https://<pre-server-dns>`
- Check the ESM agent logs: `journalctl -u psrp-esm-agent -f`

PSRP does not appear on Manage Remote Clients

- Confirm the PSRP status page shows **CONNECTING** or **PROVISIONED**
- The PRE Server administrator may need to approve the new client
- Check ESM agent logs for registration errors: `journalctl -u psrp-esm-agent --no-pager -n 50`

Resetting configuration

To start over, either:

- POST to the reset endpoint: `curl -X POST http://<vm-host>:8080/reset`
- Or SSH in and run: `sudo systemctl stop psrp-esm-agent && sudo rm /etc/psrp/config.json && sudo python3 /opt/psrp/lifecycle/manage.py reset`

SSH Access

For troubleshooting or manual inspection, SSH into the VM:

```
ssh -p 2222 psrp@<vm-host>
```

Default credentials: `psrp` / `psrp-build-temp`

> **Note**: Change the default password after deployment for production use.